MIL-E-1/739C
10 March 1969
SUPERSEDING
MIL-E-1/739B
24 September 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 2C46

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Planar triode

See figure 1

Mounting position: any

Weight: 3 ounces nominal

RATINGS: CW oscillator

Parameter:	F1	Ef	Eb	Ec	Ehk	Ib	Pp	tk	Seal temp
Unit:	MHz	V	V	V	V	mA	W	sec	°C
Maximum:	1,300	6.9	500	---	±90	40	12	---	200
Minimum:	---	5.7	---	---	---	---	---	60	---
TEST CONDITIONS:	---	6.3	250	0	---	---	---	300	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) (t = 168 hours)

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1256	Electrode current (anode)		0.65	II	Ib	6	15	mAdc
1231	Emission	(Eb = Ec)/Ik = 40 mAdc	0.65	II	Etd	---	17	Vdc
1306	Transconductance		0.65	II	Sm	1,500	5,500	μmhos
© 1266	Total grid current	Ec = -2 Vdc	0.65	II	-Ic	---	0.5	μAdc
© ---	Operation	F = 1,100 ±5 MHz; Ec = 0; Eb1 = 80 Vdc; Eb2 = 250 Vdc (see note 1) (Dwg 140-JAN)	0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		6.5	S3	If	700	900	mA
1211	Insulation of electrodes	E(g to all) = 100 Vdc E(sh to all) = -500 Vdc	6.5	S3	R	100	---	Meg
1316	Amplification factor		6.5	S3	Mu	30	80	---
1261	Electrode voltage (cutoff) (grid)	Ec/Ib = 10 μAdc	6.5	S3	-Eco	---	10	Vdc
1336	Heater-cathode leakage	Ehk ±100 Vdc	6.5	S3	Ihk	---	50	μAdc
1331	Direct interelectrode capacitance	No voltages (DESC Dwg 65025)	6.5	S3	Cgp Cin Cout Cksh	1.50 1.90 --- 100	1.90 2.45 0.025 450	pF pF pF pF
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; Eb = 300 Vdc; Rk = 200 ohms; t = 500 hrs	---	---	---	---	---	---
© ---	Life-test end point	Operation; Eb1 = 90 Vdc; Eb2 = 220 Vdc	---	---	---	---	---	---
	Periodic-check tests							
1031	High-frequency vibration	Ebb = 300 Vdc; Rp = 10,000 ohms; Ec/Ib = 6 mAdc (see notes 2 and 4)	---	---	Ep	---	100	mVac
---	Torque	No voltages (see notes 2 and 3)	---	---	---	---	---	---
---	Torque-test end point	Total grid current	---	---	-Ic	---	0.5	μAdc
---	Frequency drift	See notes 2 and 5	---	---	ΔF	---	2.5	MHz

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NOTES:

1. The tube shall oscillate continuously between the test points when Eb is raised slowly from zero.

2. Test 10 tubes selected at random from the first production lot of each calendar year, and approximately every 90 days of production during the year. If more than one tube fails to pass the specified end point, the failed test shall become a part of quality conformance inspection, part 2, on all lots in process. After three consecutive successful submissions, the test shall revert to the 10-tube quarterly test.

3. Apply torque of 20 lb-in between anode and pins without shock.

4. Vibrate in direction perpendicular to plane of grid.

5. Measure frequency under conditions "A" after tube has become stable, and again under condition "B" when tube has become stable. The difference between these two frequencies is defined as the frequency drift.

	Condition "A"	Condition "B"
Eb	135 Vdc ±2%	165 Vdc ±2%
Ef	5.7 ±0.1 V	6.9 ±0.1 V
TA	30° ±5° C	80° ±5° C

Place tube in standard frequency drift test set, CRG Model TF37AX, or equivalent.

Custodians:
Army - EL
Navy - EC
Air Force - 85

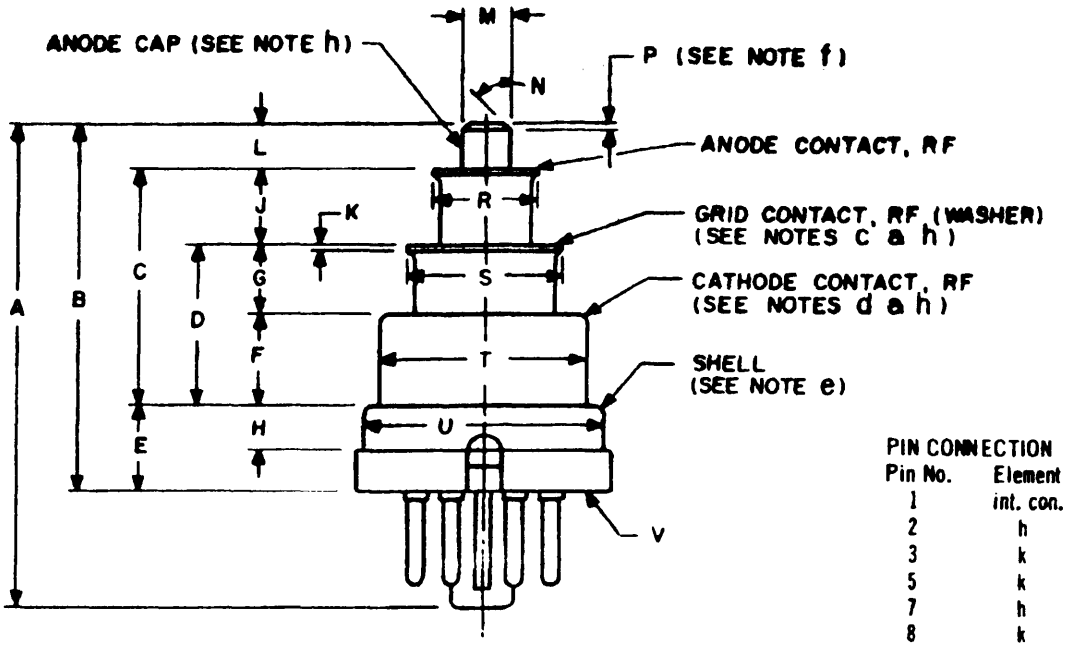
Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity
Air Force - 85

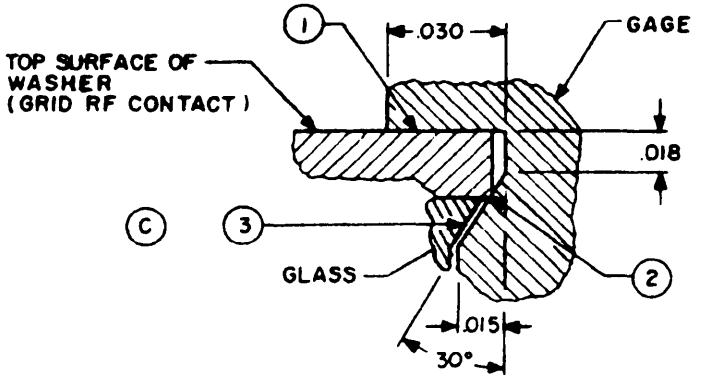
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PIN CONNECTION

Pin No.	Element
1	int. con.
2	h
3	k
5	k
7	h
8	k
cap	a



With top surface of washer (grid rf contact) in contact with surface (1) of gage and lower edge of washer (grid rf contact) in contact with surface (2) of gage, external glass surface (3) of tube shall not be in contact with gage at any point. Equivalent gaging procedure may be used.

GAGING PROCEDURE, GLASS SURFACE

FIGURE 1. Outline drawing of electron tube type 2C46.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
B		2.078 (52.78)
C	1.220 (30.99)	1.260 (32.00)
D	.850 (21.59)	.880 (22.35)
F	.475 (12.07)	.505 (12.83)
G	.360 (9.14)	.390 (9.91)
L	.347 (8.81)	.363 (9.22)
M	.371 (9.42)	.379 (9.63)
S	.808 (20.52)	.816 (20.73)
T	1.023 (25.98)	1.039 (26.39)
(see note b)		
A		2.688 (68.27)
E	.395 (10.03)	.455 (11.56)
H	.180 (4.57)	.210 (5.33)
J	.360 (9.14)	.390 (9.91)
K	.030 (.76)	.035 (.89)
P	.016 (.41)	.047 (1.19)
R	.557 (14.15)	.567 (14.40)
U	1.216 (30.89)	1.236 (31.39)
N	45° (see note a)	
V	Base: B6-108	

NOTES:
a. This dimension is for information only, and is not required for inspection purposes.
b. These dimensions shall be inspected on 10 tubes a month when in continuous production. Failure of more than one tube to meet tolerances for any dimension shall cause that dimension to become for all lots in process, part of quality conformance inspection, part 2.
c. External glass surface of tube shall conform with the detailed requirements of figure 1. Quality conformance inspection, part 2, shall apply.
d. Cathode rf contact surface shall be free from welding flash material, quality conformance inspection, part 2, shall apply.
e. Limits of dimension U do not apply at points where shell is crimped to base.
f. Radius with limits as specified may be used instead of chamfer.
g. External metal surfaces, except base pins, shall be silver plated 100 msi, minimum. Quality conformance inspection, part 2, shall apply.
h. Eccentricity of contact surfaces shall be as follows. Quality conformance inspection, part 2, shall apply.

Contact surface	TIR, maximum	Reference surface
Grid	.016	Anode cap
Cathode, rf	.016	Grid

FIGURE 1. Outline drawing of electron tube type 2C46 - Continued.

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